

MD. ROBIUL HASSAN

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate from BRAC University (CGPA 3.50/4.00, eligible for Distinction Award upon degree completion) with proven expertise in full-stack web engineering and applied machine learning research. Successfully defended undergraduate thesis developing a multimodal mental health assessment framework for rural Bangladeshi women, achieving 94.12% accuracy via multilingual text embeddings and SVM. Demonstrates strong industry software development experience across two agencies and GAOTek Inc., building high-performance web applications, RESTful microservices, and database systems. Seeking admission to a Master of Computer Science program at top-ranked universities to advance research in intelligent systems, machine learning, and scalable software architectures.

EDUCATION

Bachelor of Science in Computer Science and Engineering May 2022 – August 2026
BRAC University, Dhaka, Bangladesh CGPA: **3.50 / 4.00**

• **Academic Honor: Distinction Award — To be awarded upon degree completion (CGPA 3.50 – 3.64 tier)**

Research Interests: Applied Machine Learning, Natural Language Processing, Speech & Acoustic Analysis, Multimodal Systems, Distributed Web Engineering

Higher Secondary Certificate (HSC) — Science 2019 – 2021
Udayan Uchha Madhyamik Bidyalaya, Dhaka (Dhaka Board) GPA: **4.83 / 5.00**

UNDERGRADUATE THESIS & RESEARCH

A System to Support Mental Health of Rural Women May 2025 – August 2026
Department of Computer Science and Engineering, BRAC University (Status: **Defended**)

Supervisors: [Dr. Amitabha Chakrabarty](#) (Professor) & [Dewan Ziaul Karim](#) (Senior Lecturer)

- Formulated a machine-learning framework designed to support depression and anxiety risk assessment among rural Bangladeshi women utilizing both vocal speech and transcribed text features.
- Best Depression Model:** Achieved **94.12% accuracy**, **93.77% macro-F1**, and **1.00 ROC-AUC** using **Multilingual E5 text embeddings** with **Linear SVM**.
- Multimodal Anxiety Fusion:** Developed late-fusion architecture integrating text and speech probability predictions, attaining **91.18% accuracy**, **94.00% balanced accuracy**, and **89.67% macro-F1**.
- Acoustic Feature Extraction:** Extracted clinical vocal features using the **eGeMAPS** acoustic standard, achieving **88.24% accuracy** and **86.56% macro-F1** with **RBF SVM** on voice-only recordings.
- Pipeline Engineering:** Evaluated diverse classification benchmarks (Linear/RBF SVM, Random Forest, Gradient Boosting, KNN, Logistic Regression) and resolved clinical data class imbalances via **SMOTE**.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), TypeScript, C/C++, PHP, SQL
Machine Learning & NLP: Scikit-learn, Multilingual E5 Embeddings, eGeMAPS Acoustic Extraction, TF-IDF, SMOTE, Classification (SVM, RF, Gradient Boosting, KNN, Logistic Regression), Feature Engineering
Web & Full-Stack: React.js, Next.js, React Native, Node.js, Express.js, RESTful APIs, oRPC, HTML5, CSS3, Tailwind CSS, Vite
Databases & Tools: PostgreSQL, MongoDB, Mongoose, MySQL, Git, GitHub, Docker, Postman, VS Code, Linux
Core CS Concepts: Data Structures & Algorithms, Object-Oriented Design (OOP), Software Architecture, Operating Systems, Computer Graphics, Computer Networks

PROFESSIONAL EXPERIENCE

AlgoVerse Full Stack Web Developer	February 4, 2026 – August 10, 2026 Dhaka, Bangladesh
- Architected and engineered full-stack web and mobile applications leveraging React, Next.js, React Native, TypeScript, Node.js, and PostgreSQL/MongoDB. - Engineered robust RESTful backend APIs, complex database queries, and modular UI components, optimizing frontend loading times and data retrieval performance. - Spearheaded end-to-end development cycles across multi-client commercial platforms, handling responsive UI implementation, server-side logic, and third-party integrations.	
Zentorra Junior Full Stack Web Developer	August 7, 2025 – January 29, 2026 Dhaka, Bangladesh
- Constructed responsive client-facing web interfaces using React and modern CSS architectures, integrating with Node.js / Express backend endpoints. - Designed scalable data schemas in MongoDB with Mongoose; performed rigorous API validation and integration testing using Postman. - Collaborated in fast-paced sprint cycles to deliver functional web applications meeting strict client performance and accessibility guidelines.	
GAOTek Inc. Web Development Intern	January 22, 2024 – April 22, 2024 Remote
- Conducted enterprise web page performance optimization and responsive user-interface redesigns for high-traffic corporate portals. - Implemented frontend features and bug fixes across WordPress environments utilizing HTML5, CSS3, JavaScript, and Git version control within an international distributed team.	

SELECTED SOFTWARE & ACADEMIC PROJECTS

Bright Tutor — Tuition Booking Platform [Live Site] [GitHub]	2026
Tech Stack: Next.js, React, TypeScript, oRPC, Tailwind CSS, PLpgSQL, Docker	
- Built a comprehensive tuition management ecosystem connecting guardians, students, tutors, and administrators with role-based onboarding and Google authentication. - Implemented type-safe end-to-end communication using oRPC, robust relational schemas in PLpgSQL, and asynchronous messaging architecture.	
RestoFlow — Restaurant Management & POS [Live Site] [Video Demo] [GitHub]	CSE471 Architecture Project
Tech Stack: JavaScript, React, Node.js, Express.js, WebSockets, MongoDB, Cloudinary	
- Designed full-stack dining platform featuring QR-code table detection, bilingual menu (English/Bangla), dietary taste filtering, and live kitchen order broadcasts via WebSockets. - Engineered Role-Based Access Control (RBAC) for admins, waiters, and kitchen staff alongside an integrated POS engine supporting recursive bill splitting and discount codes.	
SelfShop — Commercial E-Commerce Platform [Live Site] [GitHub]	2025 – 2026
Tech Stack: PHP, MySQL, JavaScript, HTML5, CSS3, REST APIs	
- Engineered production commercial e-commerce platform supporting product cataloging, category management, dynamic cart operations, and responsive checkout workflows. - Developed administrative vendor and inventory management modules with automated order tracking and secure SQL data operations.	
Snowboard Rush — 3D Graphics Downhill Game	CSE423 Computer Graphics Project
Tech Stack: Python, PyOpenGL, 3D Mathematics	
- Engineered a 3D downhill endless-runner in Python with PyOpenGL; implemented custom perspective camera projection, collision detection, and continuous downhill velocity physics. - Developed procedural mountain hazards (static trees, breakable trees, boulders, flying birds), atmospheric foggy weather shaders, aerial jump/spin acrobatics, and multi-life gameplay logic.	
ACADEMIC HIGHLIGHTS & CORE COURSEWORK	
Core CS Coursework:	Data Structures, Algorithms, Software Architecture (CSE471), Computer Graphics (CSE423), Operating Systems, Database Management Systems, Artificial Intelligence, Computer Networks
Additional Work:	Wholesale B2B Platform (GitHub), Bikalpo Wholesale Grocery (bikalpo.com GitHub), DineEase Restaurant Reservation (GitHub), 3D Enemy Shooter (PyOpenGL), DIMS E-Commerce (rangpurelectrohub.com GitHub), GymFit Web Portal (GitHub), Rockdale Real Estate (GitHub), Tic-Tac-Toe (GitHub)

ACADEMIC REFERENCE

Dr. Amitabha Chakrabarty Professor, Department of Computer Science and Engineering Email: amitabha@bracu.ac.bd • Faculty Profile	Undergraduate Thesis Supervisor BRAC University, Dhaka, Bangladesh
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